ENV100 Class 5 onwards notes

Lecture 5

Why are people poor?

The poor person is one who must wait a week to see a doctor and has no social or economic clout to change the situation. They are insignificant, who might belong to a spurned race, and has been culturally marginalized, and may be socially insignificant.

Whose agenda is prioritized:

Developmentalist agenda: Northern industrialized countries defining political, economic condition

Host country: Increasing deforestation, urban industrial pollution; soil and water contamination from chemical-based agriculture

Core-periphery model

The core-periphery model is where development leads to dependence. High profit consumption goods led to cheap labor and raw materials, and that cycle repeats.

Why are people poor?

World-systems theory with greater number of players at different stages engage in exploitative, dependent economic relationships.

Sustainability

Sustainable development is defined as *meeting the needs of present without compromising the ability of future generations to meet their own needs.*

What is a "need"? Something that is necessary. Are all "needs" met now equitably? What might happen to future generations? Does it have to be all about the human?

What has been done?

The UN Framework Convention on Climate change recognizes historical and current global emissions and common but differentiated responsibilities and capabilities. 200 countries have signed but are not binding.

The Kyoto Protocol of 1997 tries to place binding targets and a timeline: reduce emissions by at least 5% below 1990 levels. It allows for emissions trading and clean development mechanisms. However, US and Australia refused to sign; US wanted cuts on developing countries.

The real need is to reduce emissions from 60-80% of the 1990 GHG levels by 2050, not 5%. However, most countries are based on self-interest and not moral principle.

What is morally correct? Criteria for moral adequacy

1. Historical responsibilities
    a. Do we have an obligation to fix the past?
2. Present capacities
    a. What can we do right now?
3. Sustainability

4. Fair process

Recall that the most developed countries tend to be the ones producing the most greenhouse gasses.

A look at inequality and environmental racism

The impoverishment of the human is embedded in, not parallel to, ecological destruction; in other words, both are expressions of the same process. Poverty, in this sense, is grounded in unequal ecologies.

Regarding environmental racism, the United Church of Christ Commission for Racial Justice found that:

- Races are the most potent variable in predicting where waste facilities sites are located
- Inequality: disproportionate burden/stress from environmental factors are placed on minority community
- Environmental justice: *Fair treatment and meaningful involvement of all people regardless of race, color, national origin, or income with respect to the development, implementation, and enforcement of environmental laws, regulations, and policies*

A look at environmental racism

- Over 30 petrochemical and industrial plants are located within a 2-mile area of Mossville.
- Two companies bordering the community of Mossville have admitted to contaminating water with millions of pounds of ethylene dichloride, a suspected human carcinogen.
- In November 1999, a US government health agency issued a report that "dioxin levels are elevated in residents of Mossville"; local exposures are at least partially responsible for these levels
- Dioxin is recognized as the most toxic substance known to science because of its wide variety of destructive effects.

Liberation

Liberation is the hope of poor and oppressed persons to achieve economic, social, cultural, and political liberation. In other words, liberation from economic, social, and political oppression. Those who seek change are not to oppress other groups or be who they once despised.

The "goal"

We're talking about regenerative sustainability.

- Deeply connected to social and economic issues as well as ecological issues
- Achieving it is a simultaneous increase in human wellbeing and environmental wellbeing, even flourishing
- This is process oriented; one that is inclusive of voices and different ways of knowing
- It is not a net zero change in progress, but a net positive!

In other words, having both liberation for people and liberation for the land, water, trees, animals, and plants have more than the sum of its parts.

Lecture 6

- 20% of the population is responsible for 75-80% of historical emissions.

- Equity cares most about fairness. An example is being harsher on those who are more privileged.
- Liberation is policy arrived at through consensus.
- What might be policies having a preferential option for the poor look like in Canada?
- (Hint: know the main argument and the supporting detail the author makes.)
- *A thing is right when it tends to preserve the integrity, stability, and beauty of the biotic community.* Aldo Leopold
- The universe is a communion of subjects, not a collection of objects. Thomas Berry

Aldo Leopold

- Scientist
- Professional game manager for U.S. governmental agencies

Centuries ago, people used to hunt and kill wolves all of the time. This caused the deer population to go up, which causes trees to run out of food. This causes deer to starve.

Aldo Leopold came up humans have unreal and improper self-understand of their role. Neither the wolf nor the mountain was an object. He advocates people return from being a conqueror to just a plain member and citizen of the biotic community.

The biotic pyramid

Humans are not on top of the pyramid. The pyramid narrows as we go up the change. Generally, only between 10% and 20% of the biomass of one trophic level is harvested by the next level. Reasoning is that you can't consume everything of an animal (such as bones).

The expansionist circle of ethics

- Center circle is the most important
- Expanding the circle means more tolerance to more people
- Moving out, these people become less important
- Why were women eventually brought into the circle of ethics? What took so long?
- What patterns can you see with the extension of universal concern for women and other minorities?
    - Accepting change takes time
- If we truly want to learn about animals, to figure out the contribution of "higher plants and animals native to [area]", how to think like a mountain, ethics must come first. Suddenly, you will see the land in a completely different matter. Berry states: consider the universe as a *communion of subjects, not a communion of objects.*
- Obligations have no meaning without a conscience.

The effect of Leopold's Land Ethic: Deep Ecology

Coined by Norwegian philosopher Arne Naess in 1973. It emphasizes the interdependent value of human and non-human life as well as the importance of the ecosystem and natural processes. The core principle is the claim that like humanity the living environment as a whole has the same right to flourish. It comprises both a scientific and spiritual dimension.

Some principles of deep ecology

1. Human and nonhuman life are independent of their usefulness.

2. Richness and diversity of life forms contribute to the realization of these values and are also values in themselves.
3. Humans have no right to reduce this richness and diversity except to satisfy vital human needs.
4. Policies must therefore be changed.

What is shallow ecology

No real paradigm change: we recycle, policies to pollution; make hybrid cars. Basically, simple tasks you would do to keep the environment okay

The anthropocentric model means humans are apex

Aside on Unequal Ecologies

There's a place full of agricultural pesticides. A company there made DDT. In the Vietnam war, Americans used it as a weapon. Something was a mess in this town. Back in the 1950s, people were saying that everything smells, there were issues about tomatoes, and one point in the 60s, a farmer got compensated because a lot of his cows were dying as they drank out of the neighbor springs.

Rachel Carson's silent spring… by the early 80s, testing was done, and they found NDMA (N-Nitroso dimethylamine), which was a dangerous chemical. Since then, people started to get concern, but it wasn't until 1989 until disaster struck, where water was filled with large quantities of MDMA.

Now, people are reliant on the piped-in water. The reason for disaster was because of very improper waste dispose. It was really interesting – this disaster, the townsfolk – those who were really advocating for justice were not the people who have been here for a long time. These are people who have recently been here. This tells you that need for jobs, that need for dependence on income… they need toe job. It is now under another company, and everything will return back to normal. The townspeople were not optimistic.

That town in India – the green revolution that entered in the 1960s – there was a chemical plant, and with chemicals and technologies they thought they would solve world hunger, but in 1984, a disaster struck. The city, and some 3.8K people died in the same night due to gasses released from the plant, and in the next two weeks, another 8K people died. Eventually, 16K-30K people died. Children were born with birth deformities. Eventually, that chemical company settled with $480M dollars and the matter is closed, never to be talked about again. But what Abou t those impoverished, those dependent on sustenance?

Chatting with – Redux

When talking about positive sustainability changes, we need to sustain ourselves. There are three stories:

1. Rachel Carson
    a. Former head of the fish and wildlife
    b. Silent Spring was written because DDT was sprayed over someone's bird…
    c. A chain of courage – an article was composed on DDT. Instead, Rachel Carson was called a communist by chemical companies. People in these chemical companies, some of them whistleblower. The chemical industries threatened to sue. Someone

did 80 lectures against. Everyone says Rachel Carson is lying, at least those in the chemical companies.

   d. Eventually, DDT was banned. People of integrity, doing what is right, in coordinating with their jobs, help the modern environment. It is all of us, acting with integrity, that helps.

2. In Ontario, $\exists \geq 50$ smog alerts. So, someone aim was to close all coal plants in Ontario. Eventually, coal fire plants were eliminated in Ontario. Now, Smog has been reduced to nothing. Part of that chain of courage and integrity – Chemists agree, and economics agree (causing too many hospitalizations, affecting the taxpayers) – so it brought many people together.
3. One June afternoon, almost 10 years, said 'you know this tree, I got to cut it down, end of life.' We said the tree was healthy. So, the neighbor got a lawyer to threaten us. This tree was a special tree. This is a boundary tree and can be protect with the Ontario forestry act. If someone cuts off the tree, they could face severe consequences. We were sued. We had to hire a lawyer. We won and got a lot of money. In Toronto, no one could cut down boundary trees without permissions. Now, city of Toronto has changed its policies. Now, this case is being taught.

These are three transformative actions from people with the ability to change. The idea is progress, not perfection. Can we make progress? Yes.

Lecture 7

What have we been led to believe is the reason for education?

- Ignorance is a solvable problem
- With enough knowledge and technology, we can manage planet Earth.
- Knowledge is increasing by implication and human goodness
    - We know that knowledge is increasing exponentially, but is human nature also becoming better? You can give instances where it is, for sure, for example, people are no longer relentlessly violent in comparison to thousands of years ago, and slavery is no longer present in the U.S., but still in the global south people are losing their land to disease. To say that it is a given is quite dubious.
- We can adequately restore that we have dismantled
    - Different disciplines are not connected to each other and bringing them together (encouraging interdisciplinary work) is very difficult. People can't understand the language of each other; it is not necessarily bad, but not talking to each other is a concern.
- The purpose of education is that of giving you the means for upward mobility and success.
    - The plain fact is that the planet does not need more "successful" people. But it does need more peacemakers, healers, restorers, storytellers, and lovers of every shape and form. People with moral courage and willing to join the fight to make the world habitable and human. These have little to do with success as our culture has defined it. (Orr)
- Our culture represents the pinnacle of human achievement
    - Not yet… issues are not free in society.

Ignorance

Science literature has been growing, and many papers are being published minute-to-minute. What really happens in science is that it is endless. The facts are important but knowing a lot of stuff does not make you a scientist. In science, knowing a lot of stuff is not a point; it is something that gets you to more ignorance. Knowledge creates ignorance – science creates more question than it answers. What did we use knowledge for? To make better ignorance.

As students turn into graduates, they know less but know more about what they know. They get extremely focused on the subject they are studying it that they forget everything else. *This is an ignorance in part of the universe we have. Why is this important?* It is the idea that technology will save us right now. Ignorance is not a solvable problem; it is part of who we are, and we can't go thinking that we have all the answers.

Low-quality ignorance is stupidity (usually from anti-vaxxers). High-quality ignorance is finding out that there are more questions than answers. Ignorance follows knowledge.

What education must be for

1. All education is environmental education (what is the most sustainable when doing…)
2. The goal of education is not mastery of subject matter, but of one's person. What person you become, not that you mastered this or that subject; when you master yourself, then the knowledge will come, and so will the wisdom.
3. Knowledge carries with it the responsibility to see that it is well used in the world. You invent something, it is your responsibility to see that it is well-used in the world. For those who invent landmines, it is very hard to see how well it is used in the world. But when making a cellphone, or even a businessperson selling a cellphone, then you have to see whether or not it is being used well in the world. Are resources being used well? Are you invoking consumerism, that people are constantly buying things they don't need?
4. We cannot say that we know something until we understand the effects of this knowledge on real people and their communities. The chemical companies who made DDT cannot say they knew something until they understood the effects of DDT on real people and the communities. This is very much like number 3, responsibility.
5. The power of examples over world.
6. The way learning occurs is as important as the content of particular courses.

These are Orr's sixth antidotes to education.

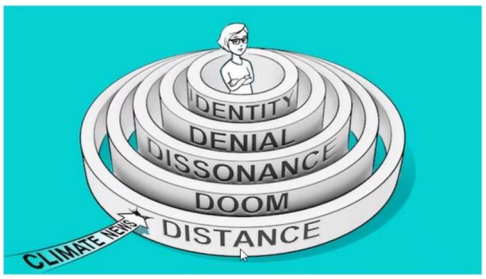
How literate students are on

- The laws of thermodynamics (energy eventually moves from a state of useful energy to useless energy.)
- The basic principles of ecology
- Carrying capacity
- Least-cost, end-use analysis
- How to live well in a place (can you live well and how can you live there)
- Limits of technology
- Sustainable agriculture and forestry
- Steady-state economics
- Environmental ethics

Why is knowledge not enough?

- Complex, wicked, or foresight problems; psychological barriers
- Selective use of knowledge for self-serving purposes
    - Confirmation bias
    - Groupthink
    - Cognitive dissonance
- Psychological obstacles to learning

We need to understand environment issues in a visceral level – in an emotional level. He says why? Because there are so much going against clear knowledge that we are dead in the water if we just rely on literacy.



As described by Per Espen Stoknes
https://www.youtube.com/watch?time_continue=854&v=dhgOebsDV7Y&feature=emb_logo

Confirmation Bias

A senator in the U.S. has a snowball in the senate and treats that as the evidence that global warming is not an issue. We are using the wrong things to confirm facts that are incorrect.

Groupthink

It is easier if everyone thinks the same. Some groups have norms. If norms are broken, then it would be a sign of weakness. Other groups, and that could be the opposite. No matter how good groups are, people must conform to what the group thinks. If a group agrees with each other, it may be commonly trench in its own correctness. *That it is more important to agree with the norm than think independently.* The problem is that an issue can be viewed in different ways, but due to groupthink, that doesn't happen.

In the U.S., it became very difficult for people to go against Trump because people were not comfortable talking about it. But for those who did, when Trump was still in office, they were ostracized. Groupthink can be very powerful. There are expected ways to think and if you break it, people can go wild.

Cognitive Dissonance

We do this all the time. It's like internal conflict. When the concepts have ethical implications, then it would be called moral dissonance. It is not knowing what is right. For example, refusing to do military service.

Most people have the moral picture of ethical people, but that does not mean they'll do everything right. Suppose you were a good person, but your boss tells you to mislead your customers. This crease cognitive dissonance, and this seems to manifest as guilt, something that you wish to resolve. Now some people will react by refusing to mislead customers, trying to preserve their self-image.

Unfortunately, most people would resolve dissonance without doing the right thing. They might think that product disinformation is not so unethical that they think, or they may rationalize that they were only doing the things that they were ordered to. Or they try to learn the product as little as possible so they could be innocently ignorance rather than intentionally honest.

We rationalize everything, but we hold to the thoughts in our heads which are conflicting causes problems and discomfort in us, and we have to do something to alleviate it. We obviously run into conflicts.

These psychological mechanisms are not evil, but they can get in the way of trying to move forward in understand climate change of environmental destruction. The suggestion is, don't get people more knowledge, help them understand the environment from a very visceral level.

Why is knowledge not enough?

We need scientific knowledge and ecological literacy, but there must also be visceral learning. Education is not enough. Love and admiration for the land is a powerful way to entice someone to not do anything unsustainable; if it is an obligation but the person following it does not know why, then they may not have an incentive to continue.

What is visceral?

"Truly," where if affects your emotions.

Hempel and Leopold were talking about developing a visceral connection to the land. Hempel used visceral, and Leopold used the words "love, respect, and admiration." From a larger lens, they are talking about the same things. These outweigh education – someone can be educated but not care because there's not enough incentive to apply what they know.

Lecture 8

Ecofeminism – Vandana Shiva. She is a physicist. She is very critical of how science is done. She is not talking as a science outsider; she is a science insider, so she knows what she is talking about.

- Woman's knowledge, women's work devalued and rendered invisible
- Nature's productive capabilities, devalued and rendered invisible
- She says women's work is rendered invisible and so are nature's protective capabilities

Science becomes very one-eyed. They tell you how to do stuff, but it tends to not talk about its consequences. We cannot say that we know something until we understand the effects of this knowledge on rea, people and their communities.

Eco-feminism

- Unites environmental ethics with feminist politics
- Dualisms – woman cognitively associated with nature
- Patriarchy – dominance of nature by man – masculine as difference honed not genetically but culturally
- It asks, whose wellbeing is being ignored.

Men are associated with the mind, abstract, and thinking. Women were cognitively associated with nature, and emotion. Patriarchy encourages men to show dominance of men over women, and because women are associated with nature.

You can have a man who is a feminist. These are *not* necessarily black-and-white male/female differences. Ecofeminism asks a central question – whose wellbeing is ignored? Eco-feminism has moved on from women but also to children and in some cases, there are feminists who bring in the LGBTQ society and other people who are being excluded – whose wellbeing is being ignored I the general patriarchal norms in society. The conclusion that would unite all eco-feminists is that the logic of domination that marginalized women is the same domination that marginalizes the world. There is a problem of seeing the nature as an object, and the same applies to women. When the patriarchal mindset connects women to nature, this is dual marginalization.

In Shiva's world, it is a recognition of the conquest of nature and the conquest oh human beings, especially women, and [I] would also add increasingly future generations, and definitely the disenfranchised everywhere. The less privileged, the less powerful are in solidarity; they are not separable. Shiva has never seen eco-feminism as an issue of essential in biology. They are not genetically wired to be closer to nature but are culturally wired because the patriarchal system pushed them out. They were biologically a woman ⇒ they are living this way, they are forced to act this way, etc. If you are from birth, you told you were going to do all the chores and the one who takes care of the kids, you are the one who will be making sure there is food and that the food is safe, and that you will have the sense of nurturing ⇒ you are not necessarily a woman. For a young boy, if you were told that, that young boy will get a very different view on the world. This is an opinion taught from lecture, and it is not guaranteed that everyone will agree with it. This is ecofeminism as Vandana Shiva expresses it.

Negative reputation

René Descartes (1596 – 1650) – I suppose the body to be just a statue, or a machine made from earth. Animals were thought to be machines. Then if a pig was kicked, was that because the pig's soul was experiencing pain, or was the pig programmed to react like that? Descartes with his thinking ushered the world with object-like thinking, and the mind and the soul were interconnected to the body, and others said to was separate.

Treating bodies as objects started the birth of this type of thinking that everyone is expendable – they are just programmed to do what they do.

Francis bacon (1561-1626) – Encourages men to enslave women. Talks to fellow scientists about that. Encourages men to be conquerors. Around the enlightenment era and the post enlightenment area, there was the birth of modern science and thinking – the idea about men being rational and women being rational.

Leopold encouraged people to stop thinking themselves as conquerors and instead themselves as just plain citizens in the biotic community.

Monoculture of the mind

- Monoculture for biofuels
- Terminator seeds
- Leads to conquest of nature, women, future, generations, disenfranchised.

Instead of planting as indigenous people where you plant multiple plants together (which helps production) – that is not monoculture in the scientific way. In the scientific way, it is just wheat. It is just one plants. Also, a monoculture of thinking – just this one way of doing things. It is that patriarchal-dominant warrior conquest that sees things in a dualistic way. The idea is terminator

seeds. For example, there is a company that decides to make a seed that once grown it will make cotton, and the plant will die yielding no seeds. This makes the farmer dependent on that company for the next yield. This is a monoculture – one way of thinking, the dominant way of thinking, the way of conquering nature, conquering women, conquering all those who are marginalized and generations thereafter. This has caused a lot of suffering to the farmers. Women still carry the culture of sharing, and caring.

If you have ideas of making money, then that is how corporations are formed.

Ecofeminism and knowledge

"The assumptions are evident: nature is unproductive; organic agriculture based on nature's cycles of renewability spells poverty; women and tribal and peasant societies embedded in nature are similarly unproductive, not because it has been demonstrated that in cooperation they produce *less* goods and services for needs, but **because it is assumed the "production" takes place only when mediated by technologies for commodity production**, even when such technologies destroy life."

Staying Alive: Women, Ecology and Development
By Vandana Shiva

If you are not actually interfering and dominating nature and manipulating nature, it has no value. Value only comes when you cut the trees and do something with the tree. The tree has no value, until you produce something with it through technology and make it to commodity. This is the monoculture of the mind she is talking about

Knowledge / Ignorance

- Reductive nature of scientific reasoning:
    - Warrior science and capitalist patriarchy – that disconnects, dissects, de-values, privatizes, and diverts 9Waters)
    - Genetic engineering (farmers seed vs. go seed)
        - Hinges the ignorance, externalizes the risks
        - Ignores whole system (Gaia approach)
        - An ocean of ignorance is created because there is an ocean of new relationships with hazards and risks.
- There is no ignorance for using a seed that has been reused for thousands of years. But the moment you used a seed that hasn't been test a lot or for very long, you have no idea what would happen. You would hide that ignorance, buy the seed, and have externalized the risk. All the risks are placed on the farmers – GMO companies wouldn't get funding if they did.

Orr's 4th principle in action

We cannot say that we know something util we understand the effects of this knowledge on real people and their communities. This can take a long time and companies do not like to wait for profits, so there is this principle called *warrior science and capitalist patriarchy.*

An example of not having to know

When I am dealing with a normal seed that farmers have, open pollinated, something that farmers can say has no risk of contamination because there is no toxin in it, I don't have to know what that seed does because it is part of the fabric of ecological reproduction and, therefore, the lack of that knowledge is not ignorance for me. I don't have to know it to do the work I do. What I have to know is the fact that this is the best seed in my field, this is the ethics of how I should share my seed with others and that is what I have to know to do the right work.

"Thinking Eco-feminism" Ramin Jahanbegloo interviewing Vandana Shiva

Using a normal seed vs. using a seed that we don't know, mainly genetically modified seeds.

As soon as you enter a genetically modified seed, you enter the ocean of ignorance. To avoid that:

- You have to know what it does to the soil organism (20% of soil organisms disappeared in 4 years of planting. That has never been studied)
- What happens to pollinators?
- What happens to us when we eat this food? Canadian scientists have found that GM crops do not break down and that it survives in bodies of pregnant women and their fetuses. It externalizes the risks.

V is making a strong case against warrior science and the capitalist system. Once in a while you get an interest public science Doi mg this little bit of work but an ocean of ignorance is created when new hazards are created.

Knowledge dictatorship – ocean of ignorance externalizes the risk by corporations. Corporations **patent** their knowledge and refuse to share it, and they hold the knowledge themselves and that is a dictatorship. V says that is wrong. At what standpoint is she looking from the world? She is looking from the particularly marginalized women, those in India who are defenseless against the wealthy and powerful corporations, and she is looking it form the point of view. She is also looking for the earth, as how eco feminism relates to how that too, is being put into jeopardy.

The ocean of ignorance is so huge that it cannot ever be filled. If you ae creating an island of knowledge and a bigger ocean of ignorance, society will reject it saying that it is too risky. This is where knowledge dictatorship comes into play, which hides the ignorance and externalizes the risks, putting the stakeholders on the victims. They externalize the risks that come to nature and people through the ignorance of those risks.

Farmers – when seeds are patented and are engineered to not yield additional seeds, it puts the fate of the famers in the hands of who sell them the seeds. If the farmers cannot reobtain these seeds for any reason, the consequences can be severe, if that happens to lots of farmers.